



Lightweight neural network for Alzheimer's disease classification using multi-slice sMRI

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ABSTRACT

Alzheimer's disease (AD) is a progressive neurodegenerative disease. Early detection and intervention are crucial in preventing the progression of AD. To achieve efficient and scalable AD auto-detection based on structural Magnetic Resonance Imaging (sMRI), a lightweight neural network using multi-slice sMRI is proposed in this paper. The backbone for feature extraction is based on ShuffleNet V1 architecture, which is effective for overcoming the limitations posed by limited sMRI data and resource-restricted devices. In addition, we incorporate Efficient Channel Attention (ECA) to capture cross-channel interaction information, enabling us to effectively enhance features of disease associated brain regions. To optimize the model, we employ both cross entropy loss and triplet loss functions to constrain the predicted probabilities to the ground-truth labels, and to ensure appropriate representation of distances between different classes in the learned features. Experimental results show that the classification accuracies of our method for AD vs. CN, AD vs. MCI, and MCI vs. CN classification tasks are 95.00%, 87.50%, and 85.62% respectively. Our method utilizes only 3.42 M parameters and 6.08G FLOPs, while maintaining a comparable level of performance compared to the other 5 latest lightweight methods. This model design is computationally efficient, allowing it to process large amounts of data quickly and accurately in a timely manner. Additionally, it has the potential to advance the intelligent detection of Alzheimer's disease on devices with limited computing capabilities.

1. Introduction

Alzheimer's disease (AD) is a degenerative disease of the nervous system that causes cognitive dysfunction, memory loss, and behavioral abnormalities. Mild Cognitive Impairment (MCI) refers to an early stage of memory or cognitive ability loss, and in certain instances MCI is an early stage of AD. In 2021, the International Alzheimer's Association estimates that 1 new patient from MCI to AD every 6 s by 2030 [1]. Although currently there is no cure or way to alter the progression of AD, early diagnosis and intervention can slow down its continuous advancement. Therefore, timely and accurate diagnosis of AD is imperative. Structural Magnetic Resonance Imaging (sMRI) is advantageous due to its high spatial resolution, which can provide detailed anatomical information of the brain [2]. This makes it a valuable tool in the diagnosis of AD. Accordingly, the use of sMRI for computer-aided technology holds great potential for practical applications.

In recent years, machine learning has emerged as a useful tool for the diagnosis of AD. Pellegrini et al. [3] employ Support Vector Machine

(SVM) and Random Forest (RF), while Kloppel et al. [4] train a linear SVM to classify AD based on T1-weight MRI images. Gray et al. [5] use RF to diagnose AD through the examination of Positron Emission Tomography (PET) and MRI images. Morra et al. [6] apply SVM and hierarchical AdaBoost models to detect AD based on MRI images. Neffati et al. [7] also improved machine learning technique through downsized kernel principal component analysis and SVM for AD classification. Although machine learning models are generally interpretable and easier to understand, their generalization performance is not always optimal and over-fitting issues are commonly encountered on new data if not properly regularized.

Compared to machine learning, deep learning methods have been found outperformed on tasks that require higher accuracy. This is due to their ability to capture complex patterns and dependencies in data. There are two main deep learning strategies for MRI feature learning. One is to extract 3D features directly from the original 3D MRI images. Farooq et al. [8] apply GoogleNet and ResNet on 3D MRI images to obtain 3D features respectively, and fuse the features to perform

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classification on AD, MCI, and Cognitively Normal (CN). A novel deep triplet network is employed by Maysam et al. [9] in the analysis of 3D MRI images and AD detection. Mehmood et al. [10] achieved an impressive accuracy of 99.05% in classifying the stage of dementia using a Siamese convolutional neural network based on 3D MRI images. For 3D data is inherently more complex and voluminous, which may lead to longer training times and increased model complexity. Hence, 3D feature learning always require more computational resources and memory. The other strategy involves slicing the 3D MRI images into sequential 2D images, and then using 2D convolution to extract features. Basher et al. [11] propose a 2D convolutional neural network (CNN), which automatically extracts 2D features and measures the volume of the left and right hippocampus to classify AD. A lightweight deep residual network for AD classification using multi-slice sMRI is proposed by Yanteng et al. [12]. Although 2D feature learning cannot fully utilize the unique spatial information of MRI, its computation is generally lower than 3D convolution, making it more suitable for devices with limited computing power [13].

Accurately identifying the Alzheimer's disease associated brain regions could improve the robustness of the model accordingly [14]. So using attention mechanism to enhance features of disease associated brain regions is necessary. The typical attention network Squeeze-and-Excitation Network (SE-Net) [15] uses SE block to compute channel-wise attention, enabling it to dynamically adjust the importance of each channel for a given feature map. Moreover, Efficient Channel Attention Network (ECA-Net) [16] employs an efficient channel attention module to reduce the loss of SE-Net dimension while capturing cross-channel interactions and enhance the feature representation of the model. Global Second-order Pooling convolutional networks (GSoP-Net) [17] can capture higher-order statistics of features by using GSoP-Net to reflect the relationship between channels in the form of covariance. The Convolutional Block Attention Module (CBAM) [18] is a type of attention mechanism module that combines spatial and channel attention to feature maps, enabling the model to selectively focus on important regions and channels.

CNNs guided by cross entropy loss emphasizes on minimizing the difference between the predicted probabilities and ground-truth labels. If the model is guided only by cross entropy loss, it may not be able to achieve stable performance due to the subtle differences in AD sMRI images, and the model may over-fitting on the training data. To mitigate these issues, additional regularization techniques or loss functions may need to be employed. Triplet loss is a type of distance metric function that is commonly used in deep metric learning. It can enhance the feature representation of the model, which can improve the distinguish ability between similar classes [19]. In addition, triplet loss can help the model learn more robust and discriminative features by enforcing that the distance between the anchor and positive sample is smaller than the distance between the anchor and negative sample. Rashid et al. [20] propose a BicephNet model which combines triplet loss and cross entropy loss to improve the performance of lightweight CNN model. Similarly, Sun et al. [21] optimize the recognition network with softmax loss and fine-tune it by adding a triplet loss to search for similar medicine images.

Medical imaging data raises concerns regarding privacy, ethical, and economic considerations, and it is often difficult to access on large scale [22]. Previous studies have shown that simple deep learning models often outperform complex models on small sample neuroimaging datasets [23,24]. Lightweight deep learning models with fewer parameters can be trained on smaller datasets and still achieve high levels of accuracy, making them more generalizable across different datasets. In this paper, a lightweight neural network using multi-slice sMRI is proposed for auxiliary classification of AD. The method consists of three main stages: (1) Converting the original 3D sMRI images into 2D slices and extracting multi-slice features through a backbone based on ShuffleNet V1 to reduce the computational demand and memory usage. (2) Using ECA to efficiently capture cross-channel interactions and enhance

the importance of task-related features. (3) Incorporating cross entropy loss and triplet loss to cluster features of the same class together, while maintaining the differentiation between features of different classes.

2. Methods

The framework of lightweight neural network using multi-slice sMRI is shown in Fig. 1: (1) Based on ShuffleNet V1, a backbone composed of multiple down sampling units and base units is used as the feature extraction module. (2) Attention module enhances the potential features based on its channel-wise importance. (3) Cross entropy loss and triplet loss are combined as the loss module of the entire work to improve the robustness of the model.

disease classification using multi-slice sMRI.

2.1. Feature extraction module

ShuffleNet V1 is a lightweight CNN design with high computational efficient and reduced parameters. It has been adapted for use with multi-slice sMRI in the proposed method. The down sampling units and base units configuration of ShuffleNet V1 is modified to (2, 4, 2), where 2 is the structure combines 1 down sampling unit and 1 base unit, while 4 is the structure combines 1 down sampling unit and 3 base units. Fig. 2 shows the down sampling unit and base unit of feature extraction module. Group convolution (GConv), Channel shuffle (CS), and Depth-wise convolution (DWConv) are used in both of the two units. GConv can reduce the computational cost by dividing the input feature maps into multiple groups, and pointwise convolution is carried out in groups accordingly. This operation is performed only within the corresponding groups. Though GConv reduces the computation, it may lead to non-circulation of information between channels [25]. Therefore, CS is introduced after the 1×1 GConv to overcome this issue. CS transposes and flattens the output of GConv in turn to evenly shuffle the channels between different groups. Then information from different channels can be mixed and combined, allowing for more interaction and information flow between channels. Furthermore, 3×3 DWConv is used to further reduce computational cost and the number of parameters while capturing spatial information. By using CS, the feature maps are restructured, and ensuring that the input of 3×3 DWConv is from diverse groups, thereby facilitating information transfer between different groups. Finally, to expand the channel dimension with minimal computation, a 3×3 AvgPool branch is added in the down sampling unit to average pool the input feature map.

2.2. Attention module

Attention networks that focus on the distribution of channel characteristics and utilize a fully connected layer to obtain channel correlation have limited efficiency in capturing cross-channel information. In contrast, ECA differs from classical channel attention mechanisms. It can effectively capture cross-channel interactions while mitigating the negative effects of dimensionality reduction. The ECA can achieve significant performance improvements with a minimal number of parameters.

The structure of ECA module can be found in the top right of Fig. 1. Tensor stitching is utilized to merge multiple slices of one subject. Then Global Average Pooling (GAP) is applied to aggregating features. For cross-channel interaction, the 1D convolution is performed on groups of 3 adjacent channels in our work. By performing the 1D convolution, local dependencies and interactions among channels can be captured. This process helps to integrate and fuse relevant features across channels, leading to a more comprehensive representation of the input data. Then Sigmoid function is applied to the output of 1D convolution to obtain the significance scores of distinct regions within the channel attention feature maps. The output of the ECA assigns weights to brain regions based on their significance scores, with disease-related regions

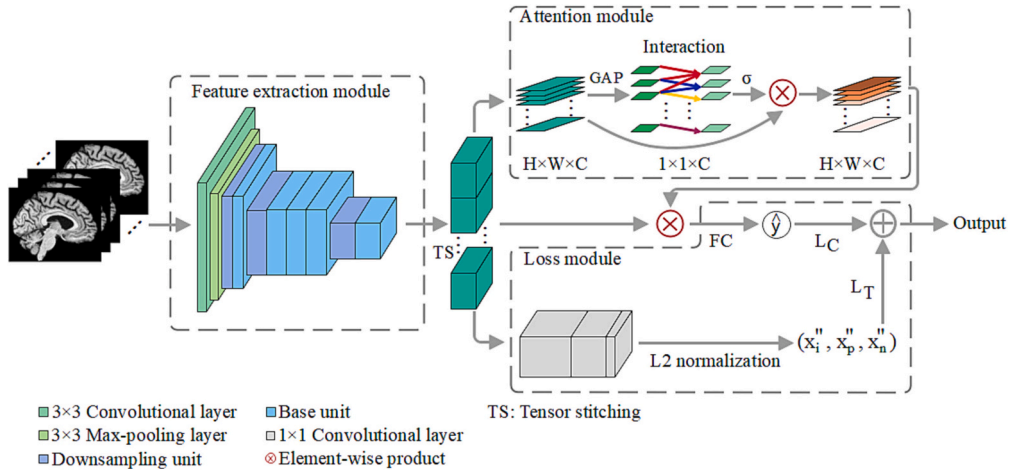


Fig. 1. Framework of the lightweight neural network for Alzheimer's.

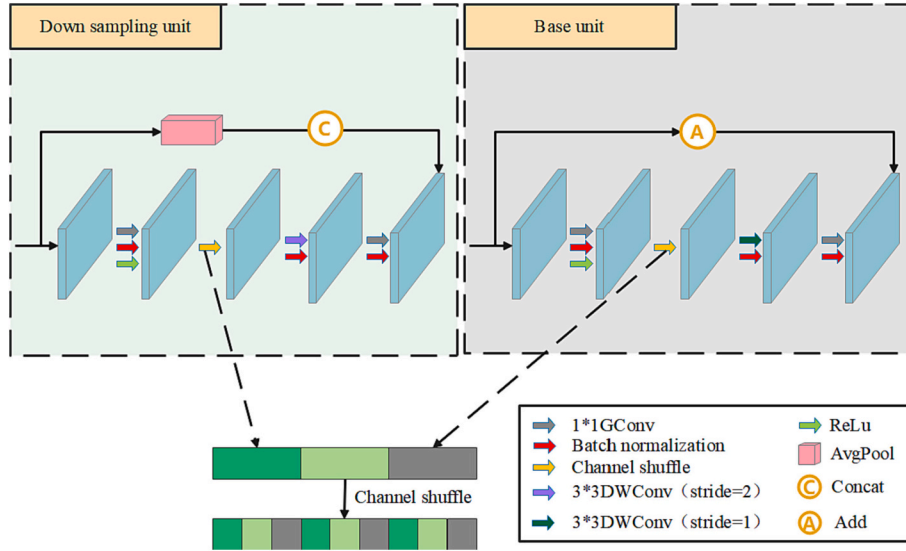


Fig. 2. Units of feature extraction module.

typically receiving greater weights. Take the output of tensor stitching as $X \in R^{W \times H \times C}$, where W , H and C are width, height and channel respectively. The weights of channels can be calculated as:

$$g(X) = \frac{1}{WH} \sum_{i=1}^W \sum_{j=1}^H X_{ij} \quad (1)$$

$$\omega = \sigma(C_1 D_k(g(X)))$$

$$x' = \omega X$$

Where $g(X)$ is channel-wise GAP for the output X . $C_1 D_k$ indicates 1 dimensional convolution with kernel size k , k also represents the coverage of 3 adjacent channels. σ is Sigmoid function, and x' is weighted feature map.

2.3. Loss module

After performing an element-wise product operation, the input to the cross entropy loss function is obtained from the output of the fully connected layer (FC), cross entropy loss can be calculated as:

$$L_C = \sum_{i=1}^{batch-size} -(y_i \log \hat{y}_i + (1 - y_i) \log(1 - \hat{y}_i)) \quad (2)$$

y is the real label of x' , $\hat{y}$ is the predicted probability value.

Triplet loss is adopted to constrain the distances of inter-class and intra-class. The loss function takes in three examples: an anchor example, a positive example (an example of the same class as the anchor), and a negative example (an example of a different class). As shown in Fig. 3, triplet loss encourages the distance between the anchor and positive example to be smaller than the distance between the anchor and negative example.

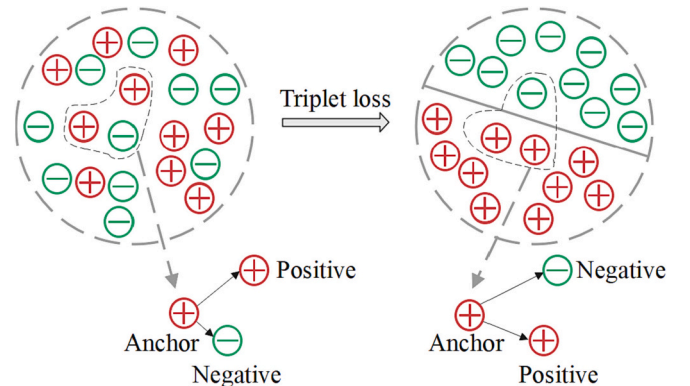


Fig. 3. The function of triplet loss module.

and negative example by a certain margin.

In another network branch, after the tensor stitching results undergo three sequential 1×1 convolution layers to reduce the channel dimensions (see Fig. 1). Subsequently, the features are then subjected to L2 normalization and used to calculate triplet loss. The triplet loss L_T is calculated as follows:

$$L_T = \sum_{i=1}^{\text{batch-size}} \max(d(x_i^a, x_p^a) - d(x_i^a, x_n^a) + \text{margin}, 0) \quad (3)$$

x_i^a , x_p^a and x_n^a are from the same batch and combined to form the triplets. In the triplets, $d(x_i^a, x_p^a)$ is the Euclidean distance of anchor example x_i^a , and positive example x_p^a which from the same class as the anchor. $d(x_i^a, x_n^a)$ is the Euclidean distance of x_i^a , and negative example x_n^a which from a different from anchor label. *Margin* is a hyperparameter that encourage the model to learn a decision boundary that maximizes the difference between the positive and negative distances.

In loss module, we consider combining cross entropy loss and triplet loss. Therefore, the loss function of the entire network can be written as follows:

$$L = L_C + L_T \quad (4)$$

3. Experiments

3.1. Data processing

Data used in this paper were obtained from the Alzheimer's Disease Neuroimaging Initiative (ADNI) database (adni.loni.usc.edu). Prior to analysis, the sMRI data underwent preprocessing steps that included Anterior Commissure-Posterior Commissure (AC-PC) correction, skull stripping and modulation into standard space using the colin27 template. In this work, a total of 1200 subjects were included, with 400 cases each for AD, MCI, and CN groups. As the sagittal plane provides detailed insights into crucial brain regions such as the hippocampus, amygdala, and corpus callosum [26], sMRI slices from the sagittal plane were utilized in this work. To minimize information loss and enhance the relevance of brain regions, 10 consecutive slices are selected from the center of each sMRI sequence. Finally, $400 \times 10 \times 3$ sMRI slices were obtained, with a slice size of 128×128 being standardized for all slices.

3.2. Contrastive models

MobileNet V2 [27] uses an innovative inverse residual structure with a linear bottleneck element, similar to a residual network but with fewer input and output channels. To fully extract features, MobileNet V2 expands the number of channels in the following process by using expansion convolution, and then reduces the output channels by using projection convolution. This approach enables parameter reduction, which is achieved by using DWConv instead of Standard convolution (SConv).

GhostNet [28] first uses SConv to generate a smaller feature map of the channel, and then uses the DWConv to further generate a new feature map. These two sets of feature maps are stitched together to obtain the final output. This approach enables the extraction of more features using fewer parameters, thus reducing the number of network parameters and computational cost. Additionally, it helps alleviate overfitting while maintaining model performance.

ShuffleNet [29] is a highly efficient convolutional neural network that leverages several techniques to significantly reduce computation costs while maintaining model accuracy. These techniques include DWConv, group convolution and channel shuffle. By using these techniques, ShuffleNet is able to achieve high accuracy with a smaller number of parameters, making it an ideal choice for resource-constrained environment where computing resources are limited.

BicephNet [30] is a lightweight framework that builds on the VGG16

architecture. The model considers the similarity between input images by employing triplet loss to group feature maps in the same class and separate feature maps in different classes. In addition, cross entropy loss is used to further enhance the model's performance in medical image classification tasks. By combining these techniques, BicephNet achieves accurate results while being computationally efficient.

3.3. Implementation details

In this work, all the experiments are implemented using the Pytorch framework and performed on Intel(R) Core(TM) i5–10,500 CPU @ 3.10GHz with 8GB RAM and NVIDIA RTX 3090 GPU. All the experiments are conducted with the following settings: trained for 100 epochs using the SGD optimizer with warmup cosine learning rate, where the initial learning rate is 0.01 and weight-decay is 0.01, the batch size is 100. To prevent over-fitting, a dropout value of 0.2 is applied to the full connection layer in our work. To evaluate the model performance, five-fold cross-validation is adopted in all the experiments.

3.4. Evaluation methods

The complexity of a model is typically measured by the number of parameters and the floating point operations per second (FLOPs). A model with fewer parameters is considered to be lighter, while a smaller FLOPs value indicates lower computational cost.

In addition, accuracy (ACC), sensitivity (SEN) and specificity (SPE) are important indicators for evaluating classification networks. In this work, we use five evaluation metrics: ACC, SEN, SPE, F1-score and area under the ROC curve (AUC). The terms TP, TN, FP and FN are used to represent true positive, true negative, false positive and false negative, respectively.

$$\begin{aligned} ACC &= \frac{TP + TN}{TP + TN + FP + FN} \\ SEN &= \frac{TP}{TP + FN} \\ SPE &= \frac{TN}{TN + FP} \\ F1 - \text{score} &= \frac{2TP}{2TP + FP + FN} \end{aligned} \quad (5)$$

4. Results and discussions

4.1. Comparison with state-of-the-art methods

In this section, we compare our method with five other methods for AD classification, which include MobileNet V2, GhostNet, ShuffleNet V1, ShuffleNet V2, BicephNet. Experiments for all methods are based on the same pre-treated multi-slice sMRI.

Table 1 shows the comparison of different methods on three measures: number of parameters, FLOPs and ACC. Our method achieves the best ACC with the lowest standard deviation. Among the compared methods, MobileNet V2 is the lightest model, with only 2.25 M parameters. But it uses many standard 1×1 convolution, resulting in 20.84G FLOPs. GhostNet gives modest performance, however its

Table 1

Parameters, FLOPs and ACC comparisons of different methods (M, G: Size of the model on disk in megabytes and Gigabyte).

Methods	Parameters(M)	FLOPs(G)	ACC(%)
MobileNet V2	2.25	20.84	82.83±7.96
GhostNet	4.23	12.97	83.00±8.33
ShuffleNet V1	3.95	10.59	87.04±4.93
ShuffleNet V2	2.50	19.83	82.46±7.80
BicephNet	29.70	4008.01	80.50±5.91
Ours	3.42	6.08	89.16±3.76

number of parameters is higher than ShuffleNet and our method. ShuffleNet uses group convolution instead of standard 1×1 convolution, which greatly reduces the number of parameters. ShuffleNet V1, ShuffleNet V2 and our method have similar numbers of parameters, 3.95 M, 2.50 M and 3.42 M respectively, with our method achieving the lowest FLOPs, only 6.08G. ShuffleNet V2 is developed from ShuffleNet V1, which is focus on reducing memory access costs. Unfortunately, ShuffleNet V2 has greater FLOPs value than ShuffleNet V1. Compare to ShuffleNet V1 which we modified as the backbone, the number of parameters in our method is 0.53 M lower. Moreover, by reducing the number of down sampling units and base units, our method's FLOPs is 4.51G lower than ShuffleNet V1. The backbone of BicephNet is VGG16, which is simple in structure but contains a great number of weights. As a result, BicephNet's parameters and FLOPs are extremely larger than the other five methods.

As shown in Table 2, for the three classification tasks, our method has outperformed MobileNet V2 by 1.50%–11.00% in ACC. MobileNet V2 is composed of a series of depthwise separable convolutions, while our method incorporates the ECA into these convolutions, which enhances the feature weights relevant to the classification task. Therefore, our method has achieved better performance compared to MobileNet V2.

GhostNet achieves ACC and F1-score of 93.87% and 94.09% in the AD vs. CN task, which are approximately 1% lower than our method respectively. GhostNet performs a series of simple linear operations on a portion of the intrinsic feature maps to generate more feature maps, which helps to reduce the computational cost of convolutions. sMRI slices of AD contain very subtle intra-slice and inter-slice features that GhostNet may not identify well, which could account for its lower performance compared to our method. Therefore, our method's structure is more suitable for sMRI images than GhostNet.

Regarding ShuffleNet V1 and ShuffleNet V2, it was found that ShuffleNet V1 had at least 1% higher ACC and F1-score than ShuffleNet V2 in all three classification tasks. Compared to ShuffleNet, our method achieved better performance by reducing the number of down sampling unit and base unit layers, adding ECA, and incorporating triplet loss. These improvements helped to enhance the feature representation and classification accuracy of our method.

BicephNet used VGG16 as its backbone for feature extraction, leading to many parameters and higher computational complexity. In contrast, our method uses fewer convolutional layers by building on the ShuffleNet V1. The experimental results have shown that our method outperforms BicephNet across all five evaluation metrics for classification. This indicates that lighter models are more effective when dealing with smaller datasets.

Table 2
Comparison of different methods on classification performance.

Tasks	Methods	ACC (%)	SEN (%)	SPE (%)	AUC (%)	F1-score (%)
AD vs. CN	MobileNet V2	93.50	93.62	93.49	98.32	93.57
	GhostNet	93.87	95.76	91.76	98.15	94.09
	ShuffleNet V1	94.00	95.07	92.86	97.94	94.16
	ShuffleNet V2	93.00	92.94	93.29	97.41	93.00
	BicephNet	88.75	81.70	96.15	96.12	88.15
	Ours	95.00	98.68	91.66	99.20	94.93
	MobileNet V2	80.62	80.48	80.76	86.71	80.98
AD vs. MCI	GhostNet	81.50	77.87	85.00	89.13	80.55
	ShuffleNet V1	83.87	81.20	87.12	93.28	83.29
	ShuffleNet V2	80.00	86.84	73.80	87.43	80.48
	BicephNet	77.50	76.62	78.31	85.60	76.62
	Ours	87.50	86.31	89.23	94.45	87.27
	MobileNet V2	74.37	73.15	76.31	82.44	73.94
	GhostNet	73.62	71.49	75.73	80.54	73.08
MCI vs. CN	ShuffleNet V1	83.25	81.14	85.11	90.35	82.84
	ShuffleNet V2	74.37	64.78	82.02	77.92	69.17
	BicephNet	75.25	77.48	73.14	81.62	75.61
	Ours	85.62	86.41	84.81	93.67	85.88

4.2. Ablation analysis and discussion

We conducted ablation experiments to evaluate the effectiveness of each module in our proposed framework. (1) Using GConv and SConv in the proposed framework (GConv+SConv). (2) Using GConv and DWConv in the proposed framework (GConv+DWConv). (3) Adding CS on the basis of (1), (GConv+CS + SConv). (4) Adding CS on the basis of (2), (Backbone). (5) Adding attention module on backbone (Backbone+ECA). (6) Loss module is added in the backbone (Backbone+Triplet). (7) Incorporating attention module and loss module into the backbone (Ours).

We conducted a comparison between DWConv associated with CS and SConv. As can be seen in Table 3, the results of GConv+SConv and GConv+CS + SConv are the same with and without CS. Unlike GConv and DWConv which both use different convolution kernels on different groups, SConv is applied on all the input channels with the same convolution kernel, so whether shuffle the channels or not has no effect on SConv. Compare GConv+DWConv and Backbone, Backbone which contains DWConv and CS achieved obviously better performance. The number of groups equals the number of channels in DWConv, non-circulation of information between channels would be happened without CS. Hence, it is necessary to use CS to help the information flowing across feature channels. The ablation experiments showed that the backbone achieved good performance, as shown in Table 3. However, the introduction of the ECA improved the accuracy of the model, and the added triplet loss can also make the performance better. While combine ECA and triplet loss can further increase the accuracy. This indicates that our method is effective and the ECA and triplet loss contribute to the improved performance. Our proposed method achieved the best performance, indicating that incorporating both ECA and triplet loss into the backbone is a promising approach for AD classification.

4.3. Visualization via CAM

To verify the effectiveness of our method in brain regions recognition, CAM [31] is used to generate the visual explanations for better understanding. We selected two subjects from each class of AD, MCI, and CN for visualization (See Fig. 4). It is evident that the location of the

Table 3
Results of ablation experiments.

Tasks	Methods	ACC (%)	SEN (%)	SPE (%)	AUC (%)	F1-score (%)
AD vs. CN	GConv+SConv	89.38	86.21	93.15	96.35	89.82
	GConv+DWConv	86.88	81.61	93.15	93.47	87.12
	GConv+CS + SConv	89.38	86.21	93.15	96.35	89.82
	Backbone	92.37	89.73	95.29	96.85	92.18
	Backbone+ECA	93.37	92.54	94.51	97.44	93.37
	Backbone+Triplet	94.00	92.59	95.54	97.74	93.93
	Ours	95.00	98.68	91.66	99.20	94.93
AD vs. MCI	GConv+SConv	85.00	82.98	87.88	93.63	86.67
	GConv+DWConv	85.00	86.17	92.10	83.33	87.10
	GConv+CS + SConv	85.00	82.98	87.88	93.63	86.67
	Backbone	86.00	80.96	91.81	93.89	85.21
	Backbone+ECA	86.37	82.26	91.31	95.01	85.74
	Backbone+Triplet	86.75	83.31	90.74	93.27	86.22
	Ours	87.50	86.31	89.23	94.45	87.27
MCI vs. CN	GConv+SConv	83.13	84.51	82.02	90.05	81.63
	GConv+DWConv	82.50	77.46	86.52	87.23	79.71
	GConv+CS + SConv	83.13	84.51	82.02	90.05	81.63
	Backbone	83.12	86.41	79.74	90.63	83.83
	Backbone+ECA	84.75	82.13	87.48	91.75	84.18
	Backbone+Triplet	84.37	87.17	81.70	92.88	84.47
	Ours	85.62	86.41	84.81	93.67	85.88

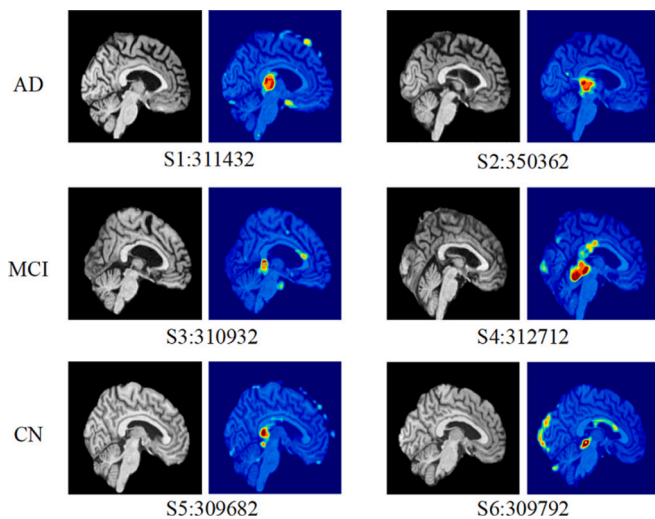


Fig. 4. CAM visualization of AD, MCI, and CN.

disease associated regions corresponds with the areas that are most commonly affected by AD, such as the hippocampus, temporal lobe and thalamus [32,33]. It verifies that the areas of interest identified by our method are reasonable and align with current knowledge of AD pathology.

4.4. Limitations

While our model achieves a better balance between classification performance and model complexity on AD, MCI, and CN, with a classification accuracy of 95%, there is room for further improvement. In the future work, we can explore the impact of cross entropy loss and triplet loss separately, and apply weights to them to further improve model performance. The feature extraction of our method is realized through 2D convolution on the sagittal plane, but incorporating interrelated feature information extracted from multi-view could help explain the intrinsic nature of AD better. In addition to slicing the sMRI data, data augmentation techniques can also be employed to overcome the limitations posed by the limited amount of available data. Take the disease-related regions directly into the model provides a viable solution to handle the high computational cost associated with processing the entire brain image. Overall, by considering the above limitations, we aim to advance the performance and interpretability of our model in future work.

5. Conclusion

In this paper, we present a novel lightweight neural network for AD classification using multi-slice sMRI. The purpose of our method is to automatically classify AD, MCI, and CN. Our proposed method achieves high classification accuracy of 95.00%, 87.5% and 85.62% on multi-slice sMRI of AD vs. CN, AD vs. MCI, and MCI vs. CN, respectively. Our method requires only a small number of parameters and FLOPs compared to other state-of-the-art methods. This is achieved by reducing the down sampling unit and base unit of ShuffleNet V1. We also added ECA to enhance the potential features of multi-slice sMRI. The loss function of the lightweight neural network is a combination of the cross entropy loss and triplet loss, which serves to constrain the classification. Experiments demonstrate that our method is helpful to improve the performance of the model, and it is expected to help improve AD intelligent detection on devices with limited computational resources.

CRedit authorship contribution statement

Qiongmin Zhang: Conceptualization, Formal analysis, Resources, Data curation, Writing – original draft, Writing – review & editing, Supervision, Project administration, Funding acquisition. **Ying Long:** Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization, Funding acquisition. **Hongshun Cai:** Software, Data curation, Supervision. **Yen-Wei Chen:** Supervision.

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